

Paper Presentation

SHAP-based Explanations are Sensitive to Feature Representation

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The Auditor's Problem

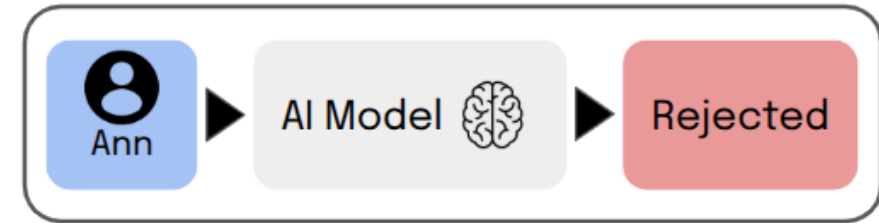
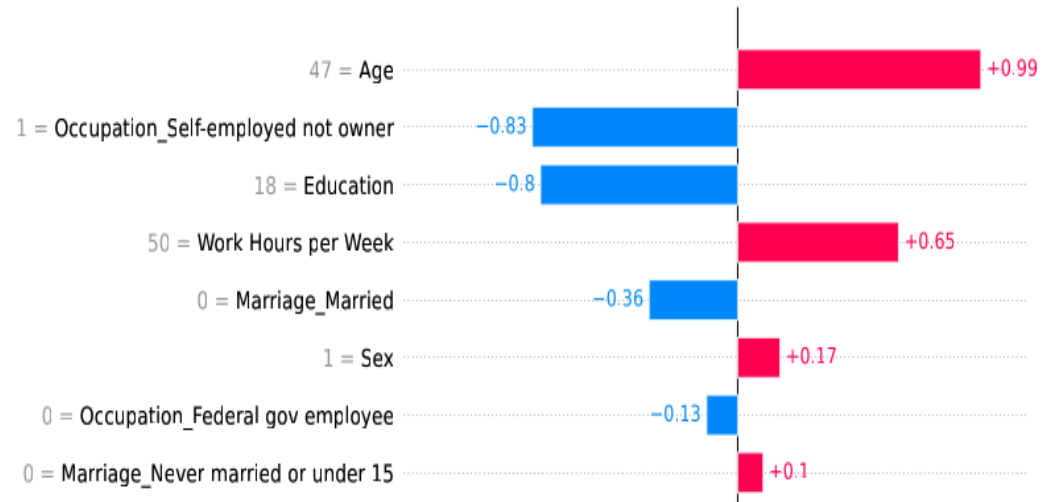
Can we trust what an AI Explanation tells us?

A bank uses an AI model to decide loan applications.

Ann is rejected.

The auditor asks:

“Did the model use Ann's age to make this decision?”



SHAP : "Age is Important"


Auditor

How much can we trust that explanation?

What If We Change Only How Age Is Represented?

Continuous Value Representation

Age : 30

Rank #1
SHAP = 0.99

Bucket Value Representation

Age bucket : 25 - 35

Rank #5
SHAP = 0.37

Note

No changes in Model or Explanation

These numbers are directly from the paper's illustrative ACS Income example: continuous age vs 12-bucket equi-width encoding .

The Core Problem

Concern

“An auditor wants to know whether an AI is using a protected attribute. SHAP seems to provide an answer. But if simply changing how that feature is represented can change its apparent importance, how reliable is that answer?”

The question

But isn't SHAP supposed to explain the model?

What happens when we change how a feature is represented? Can this be gamed?

SHAP

Shapley values quantify each feature's contribution to a model prediction.

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f_x(S \cup \{i\}) - f_x(S)]$$

Inference

Feature contribution values for a particular prediction.

The Hidden Variable

Feature Representation

The Feature is the same. The Representation is not.

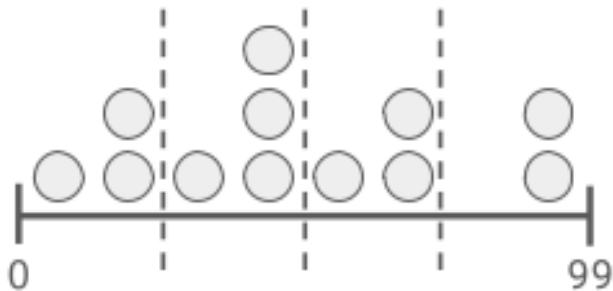
Continuous Features: Bucketization

Equi-Width

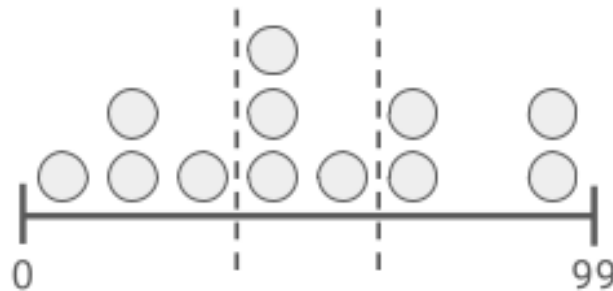
Each interval has the same numerical width.

Equi-Depth

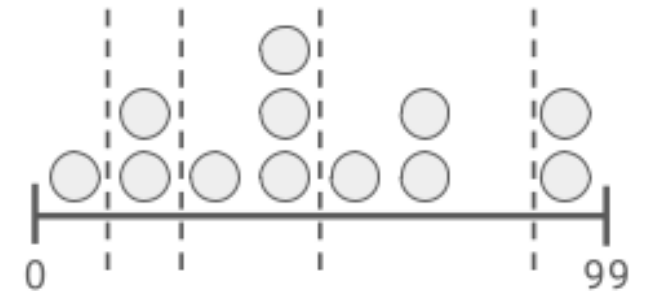
Each interval contains roughly the same number of observations.



(a) 4 equi-width buckets



(b) 3 equi-depth buckets



(c) custom buckets per Section 5.1

Categorical Features: Encoding

Original Categories

One category → one representation

We can encode categories individually, or change how categories are grouped or encoded.

Why does this matter?

Same semantic attribute → different feature representation → potentially different SHAP explanation

Strategy	Buckets
BASE	White, Black, Asian, Other
One vs. rest (OvR)	White, Rest
	Black, Rest
	Asian, Rest
	Other, Rest
2 BUCKETS	White, Black + Asian + Other
	White + Black , Asian + Other
	White + Asian, Black + Other
	White, Other
3 BUCKETS	White, Black, Asian + Other
	White, Asian, Black + Other

Why Could Bucket Size Change SHAP?

1. Bucket size changes the information available

Smaller buckets preserve more of the original feature information.

Larger buckets **merge more values together**, making the representation coarser.

2. SHAP works with that representation

SHAP measures the feature's **marginal contribution** to the prediction.

When the representation changes, the contribution being measured can also change.

From Sensitivity to Exploitability

1. How sensitive are SHAP explanations to feature engineering?

Does changing the feature representation change its apparent importance?

2. Can this sensitivity be deliberately exploited?

Can a representation be chosen to reduce the apparent importance of a protected feature?

3. If the SHAP values change, are they still faithful to the original prediction?

Definition

Fidelity : proportion of observations for which the explanation corresponds to the originally predicted outcome.

Inference from the Paper

Experiments, Results & Key Findings

The Experimental Setup

Datasets

ACS Income

46,144 observations & 8 features

ACS Public Coverage

25,524 observations & 16 features

Model

XGBoost

State-of-the-art ensemble classifier

Hyperparameter tuning
for overall accuracy

Protected Features

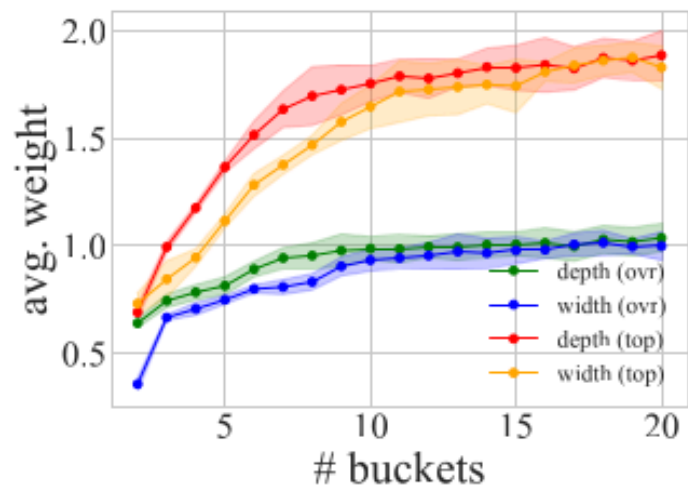
Age
Race

Both treated as protected features.

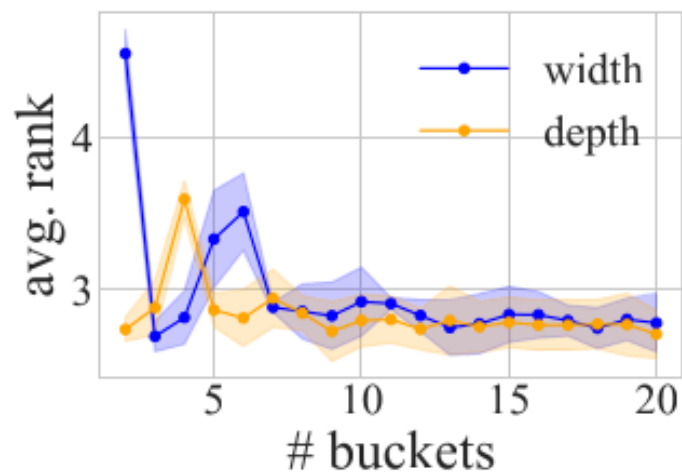
Explanation & Evaluation

- SHAP value
- SHAP rank
- Fidelity

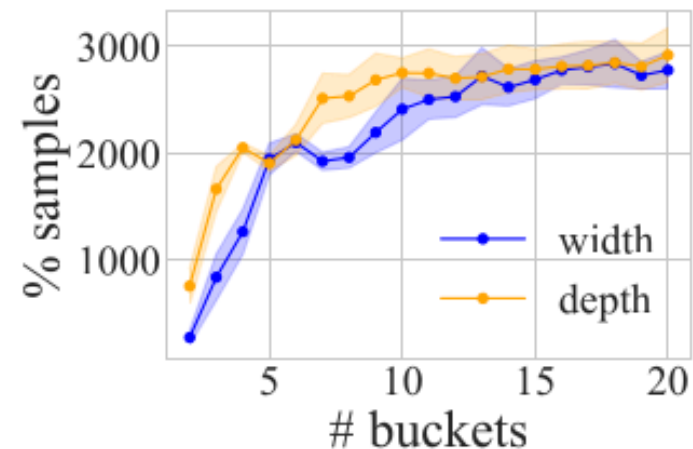
Finding #1: Age



(a) Average SHAP value for the overall population (ovr) and subset of samples with *age* as the most important feature (top).

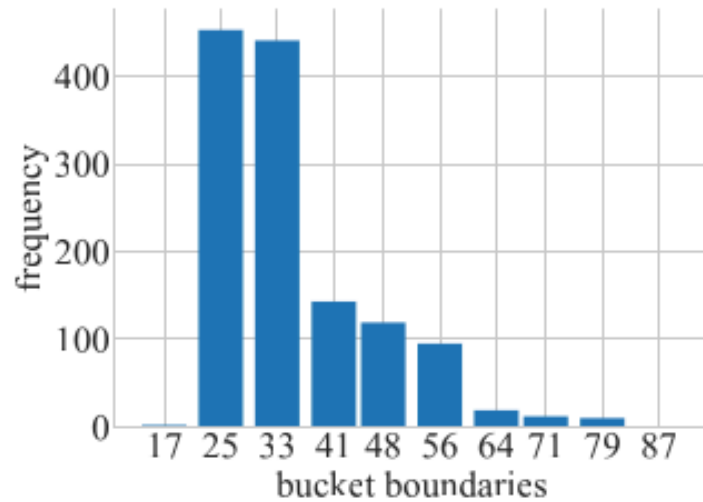


(b) Average rank of *age* in terms of feature importance.

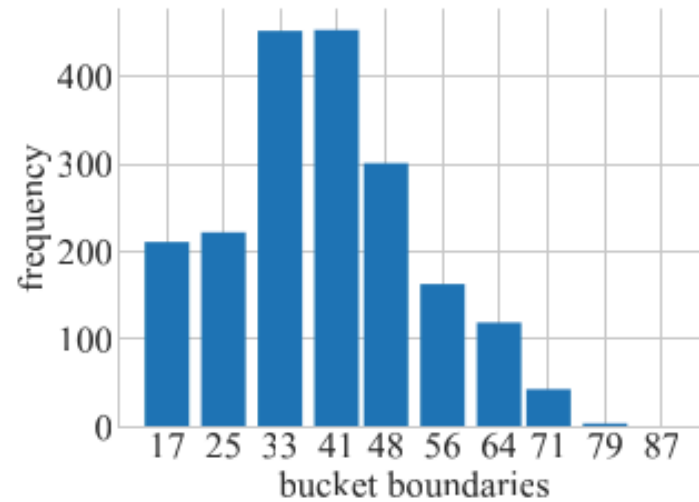


(c) Percentage of samples for which *age* is the most important feature.

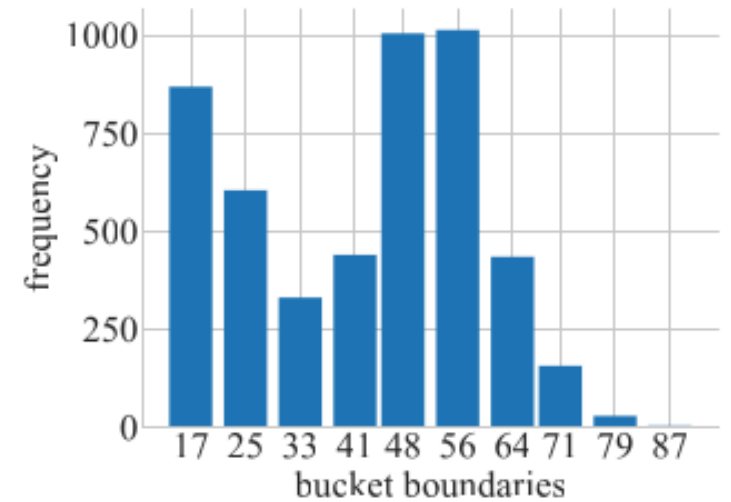
Finding #1: Age



(a) demotions of the rank of *age*



(b) promotions of the rank of *age*



(c) the rank of *age* remains unchanged

Inference

More buckets implies more importance

Finding #2: Race

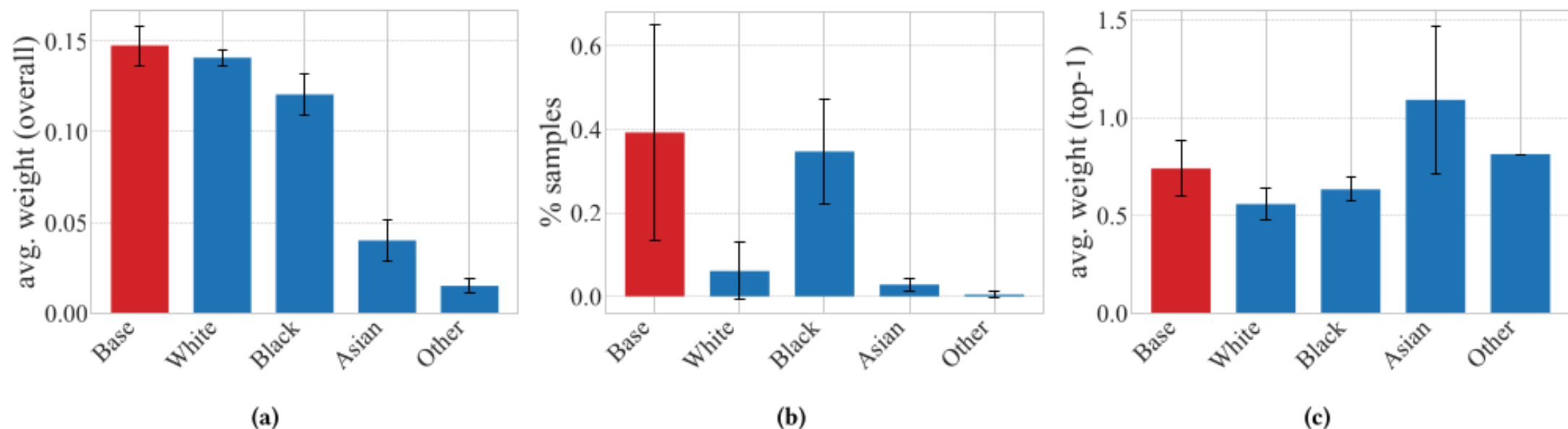


Figure 7: Effect on SHAP values using OvR bucketization for the *race* feature using the ACS Income dataset. We use each feature value (White, Asian, Black, or Other) as a category and assign the remaining values to a single category. (a) Average SHAP values of the *race* feature. (b) OvR setting versus the portion of observations where *race* is the most important feature. (c) OvR setting versus the mean SHAP value of *race* for the observations where *race* is the most important feature.

Inference

What we have seen

Changing the **feature representation** can change the SHAP explanation.

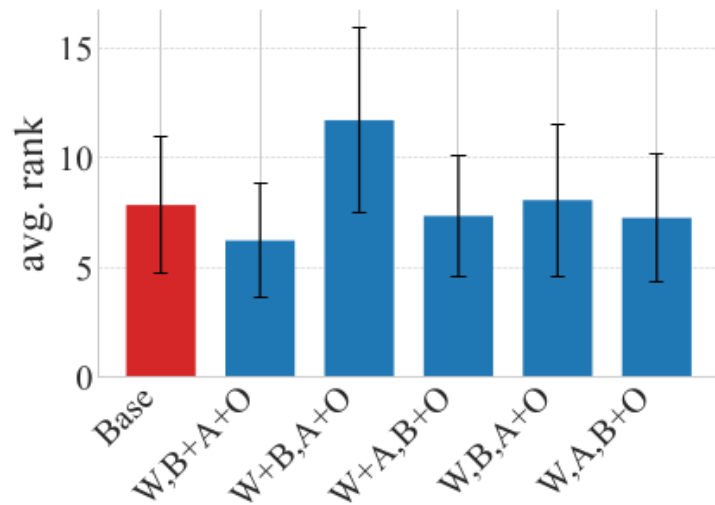
The concern

If representation affects what the auditor sees,
could someone deliberately choose a representation to hide a feature's importance?

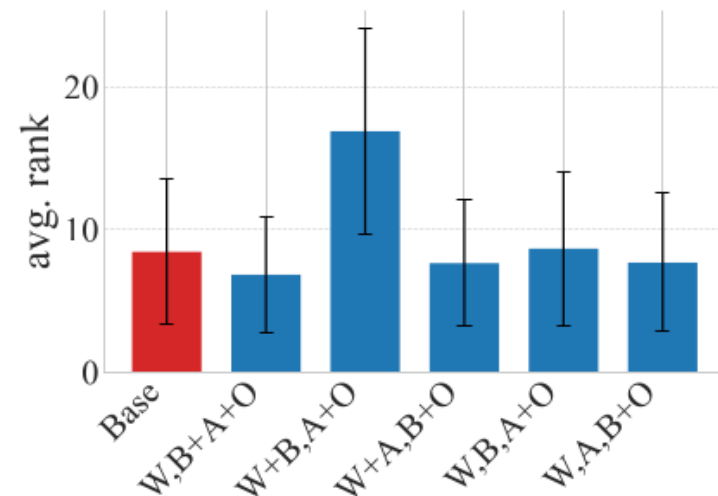
Feature-Engineering Attack

How Do You Hide a Feature From SHAP?

Bayesian Optimization Can Hide Protected Features



(a) ACS Income



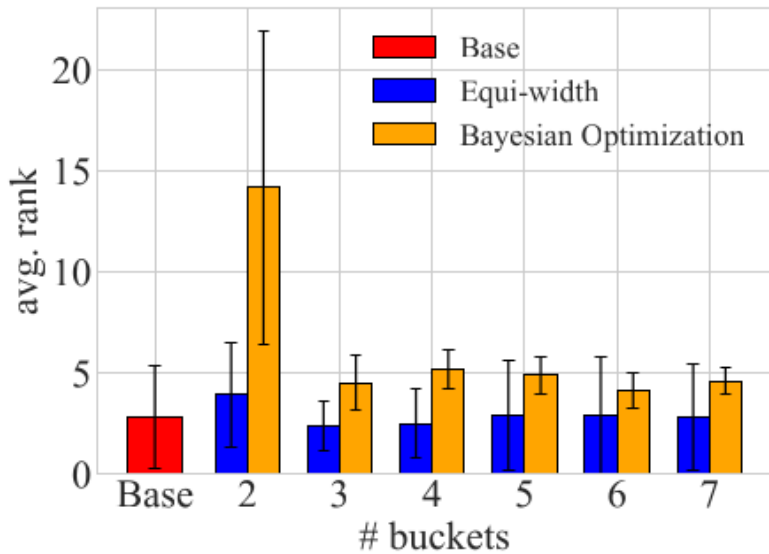
(b) ACS Public Coverage

What do we observe?

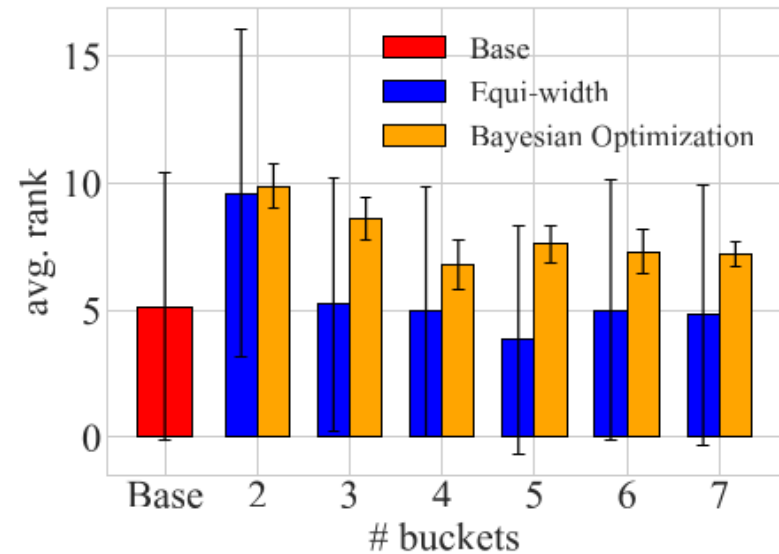
Bayesian Optimization finds representations that push the protected feature to a lower SHAP importance .

$$\min_{\tau \in \mathcal{T}} - \text{SHAP_Rank}(a, f, D) \quad \text{s.t.} \quad \lambda > \epsilon$$

Bayesian Optimization Can Hide Protected Features



(a) ACS Income ranks



(b) ACS Public Coverage ranks

Bayesian Optimization (BO) searches the representation space for a transformation that satisfies both objectives.

Final Takeaway

The Key Lesson

SHAP explanations depend on how features are represented.

What the Paper Shows

Feature representation can change SHAP importance and this sensitivity can be deliberately exploited while maintaining high fidelity.

For Auditing

Don't inspect only the model and its explanation.

The data-engineering pipeline and feature representation are also part of the explanation.

QUESTIONS ?